8 Constants

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Contents

- Constant as variables, arguments and returning types
- Constant as objects, members in a calss

8.1 Value substitution

- **The concept of *constant* (expressed by the const keyword) was created to allow the programmer to draw a line between what changes and what doesn't. This provides safety and control in a C++ programming project.**
- **It has since been put to use for *pointers*, *function arguments*, *return types*, *class objects* and *member functions*.**

8.1 Value substitution

When defining a constant, we use macro definition: #define in C. But in C++ we use a new keyword: *const*.

[*const*]() is often used when the value cannot be changed.

#define PI = 3.14 → **const double PI = 3.14;**

So programmer knows which type the PI is. With macro definition, PI is only a symbol which means 3.14 but not a double.

8.1.1 Constant value

A constant must be initialized, and cannot be asssigned to.

Constant can be defined like these:

1. **const data_type variable_name = a constant expression**
   data_type const variable_name = a constant expression

```
const int x = 10;   <---->   int const x = 10;
```

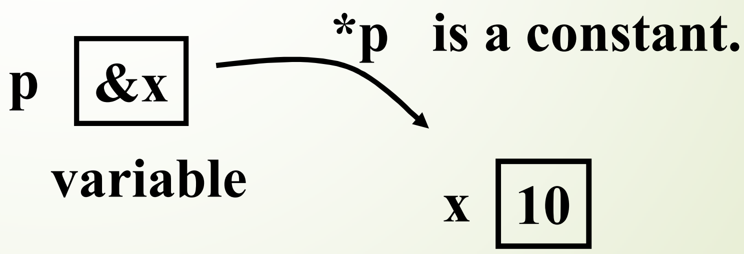
8.1.2 Pointer for constant value

2. pointer constant

   data_type const *variable_name

   In the definition, the *value of variable_name* is constant, but variable_name is variable.

```
int x = 10;
int const *p = &x;
const int *p = &x;
```



8.1.2 Pointer for constant value

```
#include <iostream>
using namespace std;
int main( )
{
        int x = 10, y = 100;
       const int *p = &x;
        // error: *p is a constant
        *p = 50;
        p = &y;     // ok
        return 0;
}
```

```
#include <iostream>
using namespace std;
double Avg(const int *p, int num)
{
        int sum = 0;
        for (int i = 0; i < num; i++)
                sum += p[i];
        return sum / num;
}
int main( )
{
        int scores[5] = {78, 82, 90, 86, 92};
        cout << Avg(scores, 5) << endl;
        return 0;
}
```

8.1.3 Constant pointer

3. pointer constant

data_type* const variable_name

In the definition, the *value of variable_name* is variable, but *variable_name* is constant.

```
int x = 10;

int * const p = &x;
```

p &x

constant

*p is a variable.

x 10

8.1.3 Constant pointer

```
#include <iostream>
using namespace std;
int main( )
{
        int x = 10, y = 100;
        int * const p = &x;

        //ok: *p is a variable
        *p = 50;
        //error: p is a constant
        p = &y;

        return 0;
}
```

```
#include <iostream>
using namespace std;
bool Update(int* const  p,  int index, int score)
{
        if ( score >100 || score < 0)  return false;

        p[index] = score;
        return true;
}
int main( )
{
        int scores[5] = {78, 82, 0, 86, 92};
        Update(scores, 2, 90);
        return 0;
}
```

8.1.4 Constant array

4. array constant

data_type const array_name[constant expression]

const data_type array_name[constant expression]

In array definition, the every element of *array_name* is constant.

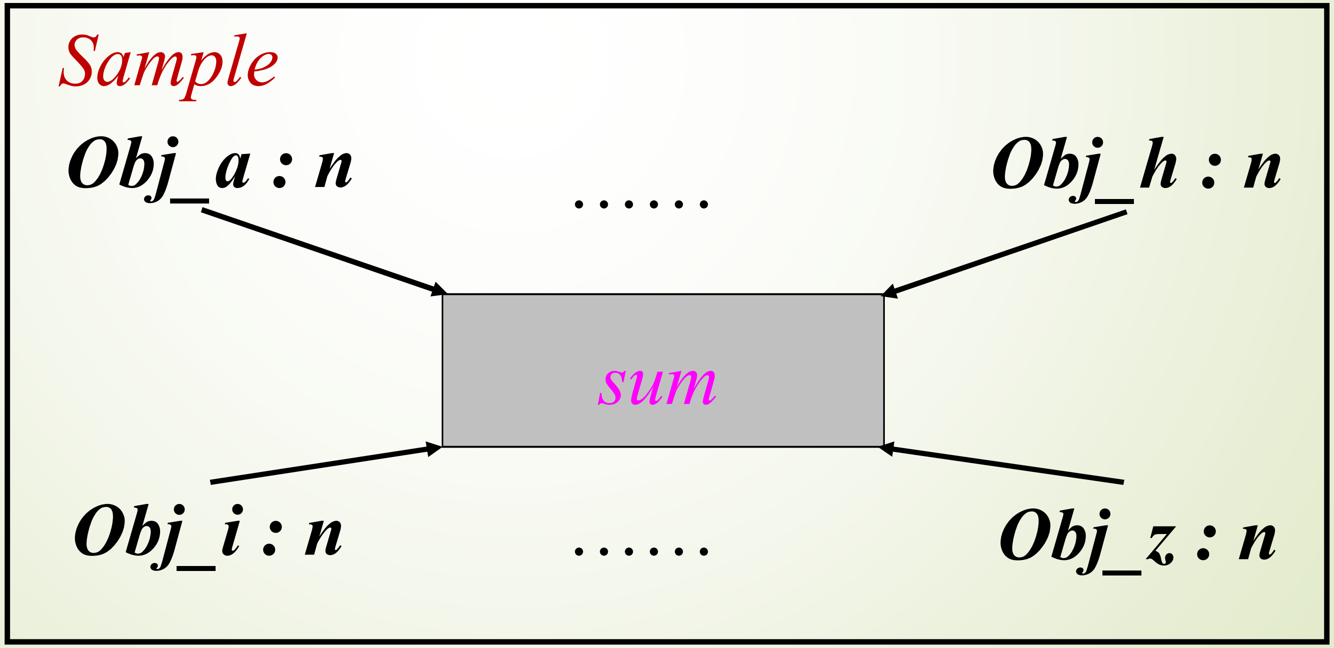
```
#include <iostream>
using namespace std;
int main( )
{
        const int a[7] = {1, 2, 3, 4, 5, 6, 7};
        a[2] = 100;     // ERROR: element is constant.
        return 0;
}
```

8.2 Classes

- Static Data Members
- Const Data Members
- Const Member Functions
- Const objects

8.2.1 static data members

```
class Sample {
    private:    static int sum;  // static data member
                int n;
} ;
```



8.2.1 Static data members

- *static* data members must be *initialized.*
  - *outside the class body*
  - *no static keyword .*
  - *qualified by the class name*

```
class Myclass
{
    static int sum;        // static data member
};

int Myclass::sum=8;  // initialization
```

8.2.1 Static data members

```
class Sample {
public:
    static void f(Sample& s)
    {
        x = 100;
        s.y = 100;
    }
    static int x;
private:
        int y;
};
// x is initialized outside of the Sample.
int Sample::x = 0;
```

Obj.x = 10; // OK!

```
int main( ) {
        Sample obj, sub;
        obj.f(sub);          //ok
        Sample::f(sub);   //ok
        Sample::x = 10;   //ok
        return 0;
}
```

Only one copy exists to all objects of the class Sample.

8.2.2 Const Data Members

- The Const data members can not be modified.
- Const data members had better be initialized in the ***member initializer list*** of constructor.
- Because ***consts and references*** must be initialized, a class containing const or reference members CANNOT be default-constructed.

8.2.2 Const Data Members

```
class A
{
public:
        A(int i);
        void Print();
private:
        const int a;        // const data member
        const int& r;       // const reference
        int    m;
};

A::A(int i):a(i),r(a)      // it must be initialized at list table
{
        m = 1;
}
```

- A class containing *const or reference* members must be initialized.
- It's best to initialize them in the member initializer list of constructor.

```
private:
   const int a = 100;
   const int& r = a;
```

member initializer list

8.2.3 Const Member Functions

data_type className::function_name (arguments) ***const*;**

```
class Date
{
 public:
      Date(int i, int j, int k) {y = i; j = m; d = k;}
      int year( )     const;
      int month( )  const  {  return m;  }
      int day( )  { return d; }
private:
      int y, m, d ;
};
```

8.2.3 Const Member Functions

- The const member functions ***do not modify*** the data members.

```
int Date::year() const
{
    return ++y; // error: attempt to change member value
}
```

- When a *const* member function is defined outside its class, the const suffix is required.

```
int Date::year() const
{
    return  y; // correct
}
```

8.2.5 mutable

```
class Date  {
 public:
        Date(int i,int j, int k) {y=i; j=m; d=k;}
        int year( )  const  { return y; }
        int month( ) const     {  return m;  }
        int day( ) const {   d++;   return d; }
private:
        int y, m;
        mutable int d;
 };
```

Summary

- The **const** keyword gives you the ability to define objects, function arguments, return values and member functions as constants.
- It can eliminate the preprocessor for value substitution without losing any preprocessor benefits.
- This provides a significant additional form of type checking and safety in your programming.